Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Contents

Wednesday 1 July 2026

1.1 Limit

Exponential Limit

Evaluate the following limit:

$$\lim_{x\to 0} \frac{(1+x)^{1/x} - e}{x}$$

Solution: To evaluate this limit, we can utilize the Maclaurin series expansion. First, let's rewrite the term $(1+x)^{1/x}$ using the exponential and natural logarithm functions:

$$(1+x)^{1/x} = \exp\left(\frac{1}{x}\ln(1+x)\right)$$

We know the standard Taylor series expansion for $\ln(1+x)$ around $x=0$:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

Substitute this back into our exponent:

$$\frac{1}{x}\ln(1+x) = 1 - \frac{x}{2} + \frac{x^2}{3} - \dots$$

Now, substitute this into the exponential function:

$$(1+x)^{1/x} = \exp\left(1 - \frac{x}{2} + \frac{x^2}{3} - \dots\right) = e\cdot\exp\left(-\frac{x}{2} + \frac{x^2}{3} - \dots\right)$$

Using the Maclaurin series for $e^u = 1 + u + \frac{u^2}{2!} + \dots$ where $u = -\frac{x}{2} + \frac{x^2}{3} - \dots$:

$$(1+x)^{1/x} = e\left(1 + \left(-\frac{x}{2} + \frac{x^2}{3}\right) + \frac{1}{2}\left(-\frac{x}{2}\right)^2 + \dots\right)$$

Expanding and keeping terms up to $O(x)$:

$$(1+x)^{1/x} = e\left(1 - \frac{x}{2} + O(x^2)\right) = e - \frac{e}{2}x + O(x^2)$$

Substitute this expansion back into the original limit expression:

$$\lim_{x\to 0} \frac{\left(e - \frac{e}{2}x + O(x^2)\right) - e}{x} = \lim_{x\to 0} \frac{-\frac{e}{2}x + O(x^2)}{x}$$

Dividing by x and taking the limit as $x \to 0$:

$$\lim_{x\to 0}\left(-\frac{e}{2} + O(x)\right) = -\frac{e}{2}$$

Thursday 2 July 2026

2.1 Limit

Algebraic Limit with Radicals

Evaluate the following limit for $k \in \mathbb{N}$:

$$\lim_{x\to\infty}\left(\sqrt{x^{2k}+x^k}-x^k\right)$$

Solution: To resolve the indeterminate form $\infty-\infty$, we multiply and divide the expression by its conjugate:

$$\lim_{x\to\infty}\frac{\left(\sqrt{x^{2k}+x^k}-x^k\right)\left(\sqrt{x^{2k}+x^k}+x^k\right)}{\sqrt{x^{2k}+x^k}+x^k}$$

Using the difference of squares $(A-B)(A+B)=A^2-B^2$ in the numerator:

$$\lim_{x\to\infty}\frac{(x^{2k}+x^k)-(x^k)^2}{\sqrt{x^{2k}+x^k}+x^k}=\lim_{x\to\infty}\frac{x^{2k}+x^k-x^{2k}}{\sqrt{x^{2k}+x^k}+x^k}=\lim_{x\to\infty}\frac{x^k}{\sqrt{x^{2k}+x^k}+x^k}$$

Factor out x^{2k} from inside the square root in the denominator:

$$\sqrt{x^{2k}+x^k}=\sqrt{x^{2k}\left(1+\frac{x^k}{x^{2k}}\right)}=x^k\sqrt{1+x^{-k}}$$

Substitute this back into the limit:

$$\lim_{x\to\infty}\frac{x^k}{x^k\sqrt{1+x^{-k}}+x^k}$$

Divide the numerator and the denominator by x^k:

$$\lim_{x\to\infty}\frac{1}{\sqrt{1+x^{-k}}+1}$$

As $x\to\infty$, the term $x^{-k}\to 0$ (since $k\in\mathbb{N}$ and therefore $k>0$). Thus, the limit evaluates to:

$$\frac{1}{\sqrt{1+0}+1}=\frac{1}{1+1}=\frac{1}{2}$$

Friday 3 July 2026

3.1 Limit

Power Series Limit

Evaluate the following limit:

$$\lim_{x\to 0}\left(-\sum_{r=1}^{\infty}\frac{(-1)^r x^{r-1}}{r}\right), \quad |x|<1$$

Solution: Let $S(x)$ denote the sum inside the parenthesis:

$$S(x) = -\sum_{r=1}^{\infty}\frac{(-1)^r x^{r-1}}{r} = \sum_{r=1}^{\infty}\frac{(-1)^{r-1} x^{r-1}}{r}$$

Notice that the familiar Maclaurin series for $\ln(1+x)$ is given by:

$$\ln(1+x) = \sum_{r=1}^{\infty}\frac{(-1)^{r-1} x^{r}}{r}, \quad \text{for } |x|<1$$

We can relate our sum $S(x)$ to this expansion by factoring out $\frac{1}{x}$ (for $x \neq 0$):

$$S(x) = \frac{1}{x}\sum_{r=1}^{\infty}\frac{(-1)^{r-1} x^{r}}{r} = \frac{\ln(1+x)}{x}$$

The problem asks for the limit as $x \to 0$:

$$\lim_{x\to 0} S(x) = \lim_{x\to 0}\frac{\ln(1+x)}{x}$$

This is a standard fundamental limit. Applying L'Hôpital's Rule (since it yields the indeterminate form $0/0$):

$$\lim_{x\to 0}\frac{\frac{d}{dx}[\ln(1+x)]}{\frac{d}{dx}[x]} = \lim_{x\to 0}\frac{\frac{1}{1+x}}{1} = \frac{1}{1+0} = 1$$

Alternatively, since the series defines a continuous function at $x=0$, we can directly evaluate the original infinite sum at $x=0$. For $x=0$, $x^{r-1}=0$ for all $r\geq 2$. The only non-zero term is when $r=1$:

$$-\frac{(-1)^1(0)^0}{1} = -(-1)(1) = 1$$

Saturday 4 July 2026

4.1 Limit

Lambert W Function Limit

Evaluate the following limit:

$$\lim_{x\to\infty} \frac{\ln(x)}{W(x)}$$

Solution: The Lambert W function, denoted as $W(x)$, is defined as the inverse function of $f(w) = we^w$. Therefore, it satisfies the defining equation:

$$x = W(x)e^{W(x)}$$

We can take the natural logarithm of both sides of this equation:

$$\ln(x) = \ln\left(W(x)e^{W(x)}\right)$$

Using logarithm properties, this expands to:

$$\ln(x) = \ln(W(x)) + \ln(e^{W(x)}) \implies \ln(x) = \ln(W(x)) + W(x)$$

Now, substitute this expression for $\ln(x)$ into the original limit:

$$\lim_{x\to\infty} \frac{\ln(x)}{W(x)} = \lim_{x\to\infty} \frac{\ln(W(x)) + W(x)}{W(x)}$$

Let $w = W(x)$. It is a known property of the Lambert W function that as $x \to \infty$, $W(x) \to \infty$. Thus, we can change the limit variable from x to w:

$$\lim_{w\to\infty} \frac{\ln(w) + w}{w}$$

Split the fraction:

$$\lim_{w\to\infty} \left(\frac{\ln(w)}{w} + \frac{w}{w}\right) = \lim_{w\to\infty} \left(\frac{\ln(w)}{w} + 1\right)$$

It is a standard result that logarithmic functions grow much slower than polynomial functions, so $\lim_{w\to\infty} \frac{\ln(w)}{w} = 0$. Therefore, the limit is:

$$0 + 1 = 1$$

Sunday 5 July 2026

5.1 Limit

Infinite Series and Product Limit

Evaluate the following limit:

$$\lim_{x\to 0} \frac{1}{x^2} \sum_{n=1}^{\infty} \ln\left(1 - \frac{x^2}{n^2\pi^2}\right)$$

Solution: Using the properties of logarithms, the infinite sum of logarithms can be rewritten as the logarithm of an infinite product:

$$\sum_{n=1}^{\infty} \ln\left(1 - \frac{x^2}{n^2\pi^2}\right) = \ln\left(\prod_{n=1}^{\infty}\left(1 - \frac{x^2}{n^2\pi^2}\right)\right)$$

We recognize the infinite product inside the logarithm as the Euler infinite product formula for the sine function:

$$\sin(x) = x\prod_{n=1}^{\infty}\left(1 - \frac{x^2}{n^2\pi^2}\right)$$

Rearranging this formula gives the value of the infinite product:

$$\prod_{n=1}^{\infty}\left(1 - \frac{x^2}{n^2\pi^2}\right) = \frac{\sin(x)}{x}$$

Substituting this back into our logarithm:

$$\sum_{n=1}^{\infty} \ln\left(1 - \frac{x^2}{n^2\pi^2}\right) = \ln\left(\frac{\sin(x)}{x}\right)$$

Now, the original limit becomes:

$$\lim_{x\to 0} \frac{\ln\left(\frac{\sin(x)}{x}\right)}{x^2}$$

To evaluate this, we use the Maclaurin series expansion for $\sin(x)$:

$$\sin(x) \approx x - \frac{x^3}{6} + O(x^5)$$

Dividing by x yields:

$$\frac{\sin(x)}{x} \approx 1 - \frac{x^2}{6} + O(x^4)$$

Now apply the Maclaurin series for $\ln(1+u) \approx u$ for small u, where $u = -\frac{x^2}{6} + O(x^4)$:

$$\ln\left(1 - \frac{x^2}{6} + O(x^4)\right) \approx -\frac{x^2}{6} + O(x^4)$$

Substitute this expansion back into the limit:

$$\lim_{x\to 0} \frac{-\frac{x^2}{6} + O(x^4)}{x^2} = \lim_{x\to 0}\left(-\frac{1}{6} + O(x^2)\right) = -\frac{1}{6}$$